

BASIC ELECTRICAL ENGINEERING

Common to All Branches

ES 301 EE

Instruction: 3+1 periods per week

CIE: 30 marks

Credits: 4

Duration of SEE: 3 hours

SEE: 70 marks

Objectives:

1. To provide an understanding of basics in Electrical circuits.
2. To provide an overview of ordinary differential equations

Outcomes: Student will be able to:

1. To analyse Electrical circuits to compute and measure the parameters of Electrical Energy
2. To comprehend the working principles of Electrical DC Machines
3. To identify and test various Electrical switchgear, single phase transformers and assess the ratings needed in given application
4. To comprehend the working principles of electrical AC machines

UNIT – I

DC Circuits: Electrical circuit elements (R, L and C), voltage and current sources, Kirchhoff current and voltage laws, analysis of simple circuits with dc excitation. Network Reduction Technique, Mesh and Nodal Analysis, Superposition, Thevenin and Norton Theorems.

UNIT – II

AC Circuits: Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C, and RL, RC, RLC combinations (series only). Three phase balanced circuits, voltage and current relations in star and delta connections.

UNIT – III

Transformers and 3-ph Induction Motors: Transformers: Electromagnetic induction, Faradays laws, statically induced emf, Lenz law, BH characteristics, ideal and practical transformer, losses and efficiency (simple Problems only), three-phase transformer connections. **Three Phase Induction motor:** Generation of rotating magnetic fields, Construction and working of a three-phase induction motor, squirrel cage IM, slip-ring IM, Applications

UNIT – IV

Single-phase induction motor and DC Machines: Single-phase induction motor: Construction and principle of operation, Capacitor start & capacitor run motor, applications. **DC Generators:** Dynamically induced emf, Fleming's Right hand and Left hand rules, Construction and principle of operation of DC generator, EMF equation, Types of DC Generators, OCC characteristics, applications.

DC Motors: principle of operation of DC Motor, Types of DC motors, applications

UNIT – V

Electrical Installations: Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing. Elementary calculations for energy consumption, power factor improvement.